Note: In homework assignments, when we are asked to "explain" or "justify", we need to provide a proof. In fact, when a math professor asks a question, they mean "provide the answer and prove that it is true." If they say "give a counterexample" they mean "give a counterexample and prove that it is a counterexample."

Problem 1. Let $A = \{\text{United States, Japan, China, India}\}$, $B = \{\text{England, Vietnam, India, Denmark}\}$ and $C = \{x \mid x \text{ is a country in Asia}\}$. List the elements of the following sets.

1. $A \cup B$
2. $(A \cap B) \backslash C$
3. $(B \cap C) \backslash A$

Are any of these disjoint from another (i.e., is their intersection empty)? Are any subsets of another?

Problem 2. Each of the following sets is defined by the "roster method." Rewrite them in "set builder" notation.

1. $\{1, -2, 4, -8, 16, -32, \dots\}$
2. $\{\dots, -4, -1, 2, 5, 8, \dots\}$
3. $\{1, 3, 6, 10, 15, 21, 28, 36, \dots\}$
4. $\{-3, -2, -1, 0, 1, 2, 3, 4, 5\}$

Problem 3.

1. Express the empty set as the intersection of two non-empty sets, or explain why this is not possible.
2. Express the empty set as the difference of two non-empty sets, or explain why this is not possible.
3. Express the empty set as the union of two non-empty sets, or explain why this is not possible.

Problem 4. We say that a subset A of $\mathbb{R}$ is **closed under addition** if for all $x, y \in A$, it's always true that $x + y \in A$. For each of the following, either explain why the set is closed under addition or provide an example to show that it is not closed under addition.

1. The set of odd natural numbers.
2. The set of even natural numbers.
3. $\left\{ \frac{1}{n^2} \mid n \in \mathbb{N} \right\}$.

Problem 5. For each of (a) through (d) below, do the following.

1. Write the statement as an English sentence that does not use the symbols $\forall$ and $\exists$.
2. Write the negation of the statement in symbolic form without using the symbol $\neg$.
3. Write the negation of the statement as an English sentence that does not use the symbols $\forall$ and $\exists$.

(Note: You do not need to ascertain the truth values for these statements.)

(a) $(\exists q \in \mathbb{Q})(q < 1/2)$.

(b) $(\forall n \in \mathbb{N})$(If $n = k^2$ for some $k \in \mathbb{N}$, then $2^n - 1$ is not prime.)

(c) $(\exists z \in \mathbb{Z})(\sqrt{2} < z < \sqrt{3})$.

(d) $(\exists r \in \mathbb{R})(r^2 - r + 41 < 0)$.

Problem 6. Write each of the following statements as an English sentence that does not use the symbols $\forall$ and $\exists$.

1. $(\exists m \in \mathbb{Z})(\forall n \in \mathbb{Z})(m > n)$.
2. $(\forall m \in \mathbb{Z})(\exists n \in \mathbb{Z})(m > n)$.
3. $(\forall m \in \mathbb{Z})(\forall n \in \mathbb{Z})(m > n)$.

Problem 7. Write the negation of each statement from the previous problem in symbolic form without using the symbol $\neg$ *and* write each negation as an English sentence.

Problem 8. Each of the following statements is false. Give a counterexample for each.

1. If A, B are sets, then $(A \cup B)\backslash B = A$.
2. If $x \in \mathbb{Z}\backslash\{0\}$, then there exists a $y \in \mathbb{Z}$ such that $xy = 1$.
3. If A, B, C are sets, then $(A \cup B)\backslash C = A \cup (B\backslash C)$.
4. If n is a natural number, then there exist $a, b \in \mathbb{N}$ such that $a^2 + b^2 = n$.

Problem 9. For each of the following sets, write out (using logical symbols) what it means for an element x to be an element of the set.

1. $(A\backslash B)\backslash C$
2. $A\backslash(B\backslash C)$
3. $(A\backslash B) \cup (A \cap C)$
4. $(A\backslash B) \cap (A\backslash C)$
5. $A\backslash(B \cup C)$

Then determine if any are equal by identifying equivalent expressions.

Problem 10 (Practice.)**.** Do not turn these in, but I recommend completing the progress checks for sections 2.3 and 2.4 in your textbook, as well as the first bit of section 5.1